13-7-26

A-P French

Waves Lec 12

Assignment #2

Problems Ch. 3 - 12, 13, 14, 15, 16

$$\ddot{x} + \gamma \dot{x} + \omega_0^2 x = 0$$

$\omega_0 = 2$ rad/s.

$\gamma = 4$ (N.s/m.kg)

$= 4\ s^{-1}$

$x(0) = 0,$

$\dot{x}(0) = 1$

$$\lambda^2 + \gamma\lambda + \omega_0^2 = 0$$

$$\lambda^2 + 4\lambda + 4 = 0$$

$$(\lambda+2)^2 = 0$$

$$\lambda_{1,2} = -2$$

Gen. Soln:

$$x(t) = c_1 e^{\lambda t} + c_2 t e^{\lambda t}$$

$$(D_t^2 + 4D_t + 4)x = 0$$

$D_t = \frac{d}{dt}$

$$(D_t+2)(D_t+2)x = 0$$

$$u$$

$$(D_t+2)u = 0 \Rightarrow$$

$$u = e^{-2t}$$

$$(D_t+2)x = u = c e^{-2t}$$

$$x = c_1 e^{-2t} + (1)te^{-2t}$$

$$x = (A+Bt)e^{-2t}$$

$x(0) = 0$

$\dot{x}(0) = 1$

$x(0) = A = 0$

$$\dot{x} = Be^{-2t} + (A+Bt)(-2e^{-2t})$$

$$= e^{-2t}(B - 2A - 2Bt)$$

$$= B(1-2t)e^{-2t}$$

$\dot{x}(0) = 1 = B \Rightarrow x = te^{-2t}$

HW $x(0) = 1,\ \dot{x}(0) = 0$

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$\dot{x}$ =

$\ddot{x}$ =

$+\beta t)\, e^{-2t}$

$= 0$

$= 1$

$A = 0$

$D e^{-2t} + (A+\beta t)(-2e^{-2t})$

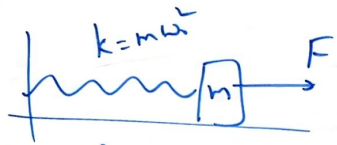
$e^{-2t}(\beta - 2A - 2\beta t)$

$\beta (1-2t) e^{-2t}$

$1 = \beta \Rightarrow x = te^{-2t}$

$x(0)=1,\ \dot{x}(0)=0$

Driven oscillations, resonance



Undamped case:

$m\ddot{x} = -kx + F_0 \cos\omega t$

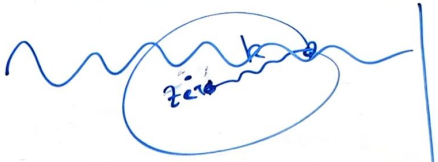
$\ddot{x} = -\omega_0^2 x + f_0 \cos\omega t$ ($f_0 = F_0/m$)

Particular sln

Choose $x = A\cos(\omega t + \alpha)$

$\dot{x} = -A\omega \sin(\omega t + \alpha)$

$\ddot{x} = -A\omega^2 \cos(\omega t + \alpha)$



$\ddot{x} + \omega_0^2 x = f_0 \cos\omega t$, $f_0 > 0$

$(-\omega^2 + \omega_0^2) A\cos(\omega t + \alpha) = f_0 \cos\omega t$

$A\cos(\omega t + \alpha) = \dfrac{f_0}{\omega_0^2 - \omega^2}\cos\omega t$, $A > 0$

For $\omega < \omega_0$,

$A = \dfrac{f_0}{\omega_0^2 - \omega^2}$,

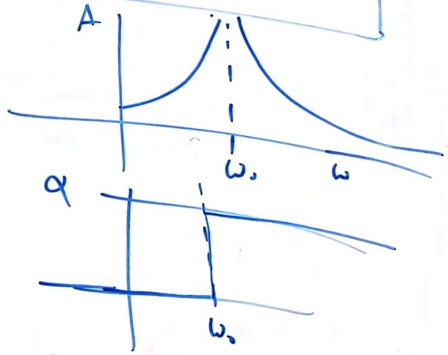
$\cos(\omega t + \alpha) = \cos\omega t \Rightarrow \alpha = 0$ (in sm osc. in ph. with driven osc.

For $\omega > \omega_0$, $A = \dfrac{f_0}{\omega^2 - \omega_0^2} > 0$

$\cos(\omega t + \alpha) = -\cos\omega t \Rightarrow \alpha = \pi$ (in sm. osc. out of phase $\Delta\varphi = \pi$ with driven osc.)

$$A = \frac{f_0}{|\omega^2 - \omega_0^2|}$$

$$\alpha = \begin{cases} 0, & \omega < \omega_0 \\ \pi, & \omega > \omega_0 \end{cases}$$



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Problems Ch. 3 - 12, 13 14, 15, 16

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$$\ddot{x} + \omega_0^2 x = f_0 \cos \omega t$$

$$+ \quad\quad\quad\quad\quad\quad \rightarrow$$

$$i\left(\ddot{y} + \omega_0^2 y\right) = i\left(f_0 \sin \omega t\right)$$

$$\ddot{z} + \omega_0^2 z = f_0 e^{i\omega t}$$

$$z = A e^{i(\omega t + \alpha)}$$

$$(-\omega^2 + \omega_0^2) A e^{i(\omega t + \alpha)} = f_0 e^{i\omega t}$$

$$(\omega_0^2 - \omega^2) A e^{i\alpha} = f_0$$

$\omega < \omega_0$,

$$A = \frac{f_0}{|\omega^2 - \omega_0^2|},$$

$$\alpha = \begin{cases} 0, & \omega < \omega_0 \\ \pi, & \omega > \omega_0 \end{cases}$$

$$z = A e^{i(\omega t + \alpha)}$$

$$x = \mathrm{Re}(z)$$

$$= A \cos(\omega t + \alpha)$$

$$= \begin{cases} A \cos \omega t, & \omega < \omega_0 \\ -A \cos \omega t, & \omega > \omega_0 \end{cases}$$

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$m\ddot{x}$

$\ddot{x}$

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$\dot{x}$

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